Electronics design notes and calculations

Source of truth: `hardware/main-board/design.py`, `hardware/led-board/design.py`, `hardware/parts.py`. Datasheet extracts with page references: `docs/datasheets/KEYSPECS.md`. Review findings and their dispositions: `docs/verification/`.

1. Power path (main board)

State	Q1 load-share (G = VBUS)	D3 S1J	Q2 load switch	Q3 rev-batt	VSYS
Battery only, switch ON	Vgs = −VBAT → on	reverse-biased (nA)	on	on (Vgs = −VBAT)	VBAT − I·(Rq1+Rq2+Rq3)
Battery only, switch OFF	on	reverse (nA)	off	on	0 (VSYS_RAW = VBAT, isolated by Q2)
USB + battery, ON	Vgs ≈ VBUS − VSYS_RAW ≈ +0.85 V → off	forward, VSYS ≈ 5.0 − F1 − 0.85 ≈ 3.9–4.1 V	on	on	≈ 3.9–4.1 V from USB (F1 PTC drop 0.03–0.1 V + D3); charger sees only the cell → terminates normally
USB, switch OFF	off	forward	off	on	0; charger runs (red LED)
Battery reversed, no USB	—	—	—	Vgs ≈ 0 → off, body diode reverse → blocks	0
Battery reversed + USB	—	—	—	charger biases Q3 on: ~42 mA precharge into the reversed cell until its PCM cuts	not blocked (round1 m2, accepted; check polarity first, VERIFY E1)

- USB input: J1 → **F1 PTC** (Bourns MF-MSMF110/16X, 1.1 A hold / 2.2 A trip, 16 V, 0.06–0.20 Ω) → D1 TVS SMF5.0A → VBUS. F1 limits a short in the neck cable while on USB. The worst-case normal draw is about 0.5 A, against a hold current of about 0.8 A at 60 °C (round2 R2-E3; the 0.75 A part had too little margin when warm). **This is not over-voltage protection** (round2 R2-E4). A 6.4–7.5 V source sits below the TVS's conduction and trips nothing, which puts VSYS at 5.6–6.6 V, above the ATtiny's and the LM75B's 6 V absolute maximum. A hard over-voltage fault destroys the TVS before or as the PTC trips. Accepted for a USB-C 5 V-only sink, with real OVP (e.g. a TPS2595-class eFuse) on the v0.2 list.
- **Q4 LED-board switch** (DMG2305UX, gate Q4_G driven from PB2 through R13 10 k, with C8 100 nF gate–drain and R12 100 k pull-up to VSYS): off in reset and in the empty-battery sleep, so a flat cell is not drained by the TLC59116. VLED = VSYS − I·R_DS(on) (≈ 10–20 mV at 180 mA). **Soft turn-on** (round2 R2-E1): R13 into C8 holds the Miller plateau, so VLED ramps at about 2.5 V/ms. Charging the LED board's ~9 µF then takes about 25 mA, where the unslowed gate gave multi-amp peaks that sagged VSYS, the MCU's supply, below its BOD level. Bench check: VERIFY E13. The I²C pull-ups (R8/R9 3.3 k) sit on VLED, so an unpowered LED board is never back-fed through SDA/SCL.
- **J4 pin 1 is VREF** (VSYS through R11 1 k) for the programmer's level sense only: never power the board from a programmer (round1 M3).

- Charge current: I = 1000 V / R_PROG = 1000 / 3.9 kΩ = **256 mA** (MCP73831 DS Eq. 5-1) ≈ 0.21 C for 1200 mAh (≈ 5 h to full). Lowered from 417 mA in round 1 (M4, m4): it keeps USB draw ≤ 0.5 A with the lamp on in most modes, and the charger dissipation ≤ ~0.7 W next to the pouch. MCP73831-2ACI/OT: 4.20 V ±0.75 %, 10 % preconditioning below 66.5 % V_REG, 7.5 % termination (31 mA), recharge at 96.5 % (~4.05 V). Thermal regulation in the charger limits current if the SOT-23-5 gets hot (worst case, IREG +10 % and V_PTH 2.79 V: (5.25 − 2.79) V × 0.282 A ≈ 0.69 W at the start of fast charge). No NTC input: charge at room temperature (user note; v0.2 MCP73833 + NTC).
- Q1–Q4 DMG2305UX: R_DS(on) ≤ 52 mΩ @ −4.5 V, ≤ 100 mΩ @ −2.5 V. The LED path runs through four FETs (Q3, Q1, Q2, Q4), then J3 and 2 × 200 mm of 28 AWG: about 4 × 0.075 Ω + 0.15 Ω ≈ 0.45 Ω, so ~80 mV at 180 mA. V_GS max ±8 V (worst case here 5.0 V).
- Q2 soft start: R7 10 kΩ into C5 100 nF (gate–source) → τ ≈ 1 ms. Q2 now charges only the main board's ~8 µF; the LED board's 9 µF comes later, through Q4's soft turn-on. Round 1 simulated 0.7–1.3 A peak for ~0.1 ms into 14–21 µF (C5 is G–S, so there is no Miller dv/dt control), which is inside the FET and connector ratings.
- Q1 gate pull-down R5 **10 kΩ** (was 100 k): after USB unplug VBUS decays with τ ≈ 70 ms, so the load spends < 0.1 s on Q1's body diode (round1 m1). Costs 0.5 mA from VBUS only while USB is present.
- Standby drain with the switch OFF (battery only): Q2 I_DSS ≤ 1 µA, D3 reverse leakage (silicon, 4 V) ≪ 1 µA, MCP73831 output reverse leakage ≤ 2 µA (I_DISCHARGE), R5 sees 0 V, R6 sees 0 V. **Total ≤ ~3 µA** plus the cell's own PCM (~3–6 µA typical): about 0.1 % of capacity per month.

2. LED drive (LED board)

- TLC59116: I_OUT = (1.26 V / R_EXT) × 15 × CG; CG = 0.992 after power-up → R_EXT 750 Ω gives **25.0 mA**/channel (TI DS Eq. 3–5). Output knee ≈ 0.6 V at 26 mA (DS: I_O specified at V_O = 0.6 V).
- Headroom: V_LED − V_f ≥ knee (≈ 0.35–0.6 V) for full current. White (Bridgelux, ~2.75 V at 25 mA): regulated down to V_LED ≈ 3.35 V (the whole usable battery range). Red (Würth, ~2.1 V): always. **Green/blue (Würth, 3.2–3.6 V at 20–25 mA)**: at 25 mA they regulate only above V_LED ≈ 3.7–3.9 V. Therefore the **Green and Blue modes run at half current** (12.5 mA, IREF step 0xBF, at every level including 15; drive 0.496 and a host test since round2 R2-F1): lower VF and knee, regulated down to about 3.6 V of battery (round1 M1). The mixed whites (3000/4000/6500 K, candle) use only a little G/B at up to 25 mA; their tint drifts slightly warm/ magenta in the last ~30 % of the battery. Accepted for v0.1 (a boost converter would add heat in the moon; v0.2 option: 5 V boost for the RGB anodes only). Bench check: VERIFY E9.
- Low-end dimming: 8-bit PWM at 97 kHz per channel × IREF current-gain steps 0.33 / 0.50 / 0.99 (8.3 / 12.5 / 25.0 mA, all inside the datasheet's 5–120 mA output ranges) → ~750:1 range. Step choice: `cg_select()` in the core (host-tested: never clips, lowest step that fits, hysteresis). Below ~2 % the minor channels of mixed modes quantise (soft orange becomes red-orange at its lowest levels: accepted, round1 F9).
- Supply current of the TLC59116 is ≤ 29 mA (max, REXT 720–750 Ω row of DS §7.5) while on, and it dominates the night-light budget (VERIFY E7).
- **Thermal protection, three layers** (round1 M-1, F1, F4):

    1. Firmware derating on the LM75B reading: 100 % ≤ 45 °C → 30 % at 52 °C → 5 % at 58 °C. Slew ~10 %/s.
    2. Current caps: 200 mA total while the sensor is *trusted* (answering, −20…100 °C, and changing within 5 min of the load exceeding 50 mA); otherwise 110 mA, and 30 % if it stops answering for 3 s. The largest mode draws 177.5 mA, so in practice the 200 mA cap never binds and the 110 mA one does. Cap changes fade over 160 ms.
    3. Hardware thermostat: LM75B OS (comparator, active low, Tos 60 °C / Thyst 52 °C, set at every init) holds the TLC59116 RESET low. Works with a hung MCU or a dead bus. LM75B POR defaults (80/75 °C) are a looser backstop until the firmware programs it.
- Recovery: 2 s watchdog; MODE1 read-back every 1 s (a POR'd TLC re-initialises). After 4 failed inits the firmware sends the TLC software reset (SWRST 0x6B: A5h 5Ah), then power-cycles the LED board through Q4, unless the thermostat is what holds it off. During the off time it drives SDA/SCL low, so VLED bleeds through the 3.3 k pull-ups (τ ≈ 16 ms, 200 ms off) to below the TLC's 0.2 V POR threshold (round2 R2-E2). RESET_N has a 2.4 k pull-up (stiffer node near the metal neck, round2 R2-E5).

3. Runtime estimates (1200 mAh, ~4.4 Wh usable to 3.3 V)

Use	Battery current (approx.)	Runtime
4000 K at full (177.5 mA LEDs at 25.0 mA/ch + ~30 mA electronics; derating may trim it once warm)	~210 mA	~5.5 h
Soft white at level 10 (~25 % of full)	~70 mA	~16 h
Red night light, level 4	~30 mA (driver-dominated)	~40 h
Empty-battery sleep (LED board off via Q4, MCU power-down + PIT)	≤ ~30 µA (VERIFY F4)	months before the PCM cut-off
Switched off	≤ ~10 µA incl. cell PCM	years (self-discharge dominates)

4. I2C

Main board pull-ups 3.3 kΩ to VLED (switched VSYS); bus = MCU + 200 mm cable + TLC59116 (0x60) + LM75B (0x48); 100 kHz. On boot the firmware clocks SCL until SDA is free: a reset mid-transfer can leave a slave holding SDA until its own time-out (TLC59116 ~25 ms), and the recovery costs nothing. Rise time with C ≈ 60–100 pF: 0.85·R·C ≈ 0.2–0.3 µs (Standard mode limit 1 µs). TLC59116 V_IH = 0.7 × VCC. The MCU runs on VSYS, and the LED board and pull-ups on VLED ≈ VSYS − 20 mV, so the levels match.